

Cubic Base Case Report

Polynomial

$$f(q) = q^3$$

Naive

$$g_{\text{naive}}(X) = X^3$$

$$\mathbb{E}[g_{\text{naive}}(X)] = q \left(6 \frac{\Delta^2}{\epsilon} + q^2 \right)$$

$$\text{Var}(g_{\text{naive}}(X)) = 18 \frac{\Delta^2}{\epsilon} \left(40 \frac{\Delta^4}{\epsilon} + 18 \frac{\Delta^2}{\epsilon} q^2 + q^4 \right)$$

$$\text{MSE}(g_{\text{naive}}) = 18 \frac{\Delta^2}{\epsilon} \left(40 \frac{\Delta^4}{\epsilon} + 20 \frac{\Delta^2}{\epsilon} q^2 + q^4 \right)$$

$$\text{Bias}(g_{\text{naive}}) = 6 \frac{\Delta^2}{\epsilon} q$$

Unbiased

$$g_{\text{unb}}(X) = X \left(X^2 - 6 \frac{\Delta^2}{\epsilon} \right)$$

$$\mathbb{E}[g_{\text{unb}}(X)] = q^3$$

$$\text{Var}(g_{\text{unb}}(X)) = 18 \frac{\Delta^2}{\epsilon} \left(28 \frac{\Delta^4}{\epsilon} + 14 \frac{\Delta^2}{\epsilon} q^2 + q^4 \right)$$

$$\text{MSE}(g_{\text{unb}}) = 18 \frac{\Delta^2}{\epsilon} \left(28 \frac{\Delta^4}{\epsilon} + 14 \frac{\Delta^2}{\epsilon} q^2 + q^4 \right)$$

Comparison

$$\text{mean gap} = -6 \frac{\Delta^2}{\epsilon} q$$

$$\text{variance gap} = -72 \frac{\Delta^4}{\epsilon} \left(3 \frac{\Delta^2}{\epsilon} + q^2 \right)$$

$$\text{MSE gap} = -108 \frac{\Delta^4}{\epsilon} \left(2 \frac{\Delta^2}{\epsilon} + q^2 \right)$$

$$\text{Bias}(g_{\text{naive}})^2 = 36 \frac{\Delta^4}{\epsilon} q^2$$